

LINDESNES FYR, Norway

WMO: 014360

Lat: 57.983N

Lon: 7.048E

Elev: 43

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	19.5	23.1	9.5	9.0	22.0	13.7	11.1	26.0	48.3	41.2	44.7	34.7	19.4	50	0.890

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.1	70.6	62.6	68.4	61.6	66.4	60.7	64.3	68.9	62.8	66.8	61.5	65.3	13.0	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.2	83.8	66.8	60.9	79.9	65.5	59.5	76.1	64.2	29.5	69.2	28.4	67.0	27.5	65.6	69.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
43.8	39.0	35.3	DB	19.7	73.3	4.6	3.2	16.4	75.6	13.8	77.4	11.2	79.2	7.9	81.5
			WB	17.7	65.6	4.3	2.3	14.6	67.3	12.1	68.6	9.7	69.9	6.6	71.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.7	36.4	35.0	38.1	43.3	50.9	55.8	60.8	60.7	56.7	50.1	43.5	39.6
	DBStd	10.34	6.14	5.93	5.03	4.02	5.40	4.29	4.11	2.80	3.34	4.25	6.06	6.73
	HDD50	2040	422	419	370	204	49	2	0	0	1	51	200	322
	HDD65	6350	887	839	835	650	438	277	145	136	249	463	644	787
	CDD50	1188	0	0	0	4	78	177	335	333	202	53	6	0
	CDD65	21	0	0	0	0	1	2	14	4	0	0	0	0
	CDH74	9	0	0	0	0	0	0	8	1	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	17.8	21.1	18.5	17.7	15.2	15.3	16.3	15.9	16.6	18.3	19.2	20.0	20.0
	PrecAvg	51.2	5.6	4.3	3.1	3.0	2.6	3.1	3.5	4.6	5.0	5.8	5.4	5.3
(15) Precipitation	PrecMax	65.3	11.6	8.1	7.0	5.2	5.4	6.5	8.2	8.2	9.0	9.8	11.2	8.3
	PrecMin	30.7	0.7	0.6	0.6	1.2	0.5	0.3	0.6	0.9	2.0	1.9	1.6	1.2
	PrecStd	8.3	2.2	2.2	1.5	1.1	1.3	1.7	1.8	1.8	2.0	2.4	2.1	2.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.3	46.0	49.5	56.0	69.5	70.6	74.7	71.4	66.4	59.9	54.9	50.9
		MCWB	46.0	44.0	43.7	47.9	61.0	59.3	65.6	62.8	61.4	56.2	52.8	48.8
	2%	DB	46.6	44.7	46.4	53.6	65.8	67.8	72.0	68.4	64.1	58.2	53.8	49.5
		MCWB	44.6	42.6	41.4	47.8	57.7	58.8	63.9	62.2	59.9	55.3	51.7	47.5
	5%	DB	45.5	43.7	44.7	51.6	62.4	65.2	69.7	66.7	62.7	56.9	52.3	48.6
		MCWB	43.6	41.9	41.9	46.4	55.1	58.2	63.0	61.4	58.9	54.0	50.0	46.7
	10%	DB	44.4	42.6	43.6	49.2	58.9	62.3	67.2	65.1	61.6	55.8	51.0	47.5
		MCWB	42.6	41.1	41.3	45.0	53.0	56.8	61.9	60.6	58.0	53.1	48.6	45.5
	0.4%	WB	46.5	44.6	45.2	50.6	62.0	62.4	67.4	65.1	62.4	57.8	53.7	49.3
		MCDB	47.7	45.3	47.7	53.8	68.6	67.3	73.5	68.4	64.4	59.2	54.6	50.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	44.9	43.2	43.6	48.5	58.5	60.2	65.2	63.5	60.9	56.2	52.1	48.1
		MCDB	46.1	44.1	45.0	51.9	64.9	65.5	70.4	66.8	63.1	57.7	53.6	49.1
	5%	WB	43.9	42.3	42.6	47.1	55.8	58.8	63.7	62.3	59.8	54.8	50.3	47.0
		MCDB	45.2	43.2	43.9	50.5	60.8	63.6	68.7	65.5	62.3	56.4	51.9	48.2
	10%	WB	42.9	41.2	41.8	45.5	53.7	57.3	62.2	61.1	58.6	53.7	49.1	45.8
		MCDB	44.4	42.4	43.2	48.7	58.1	61.6	66.6	64.4	61.4	55.4	50.8	47.2
	5% DB	MDBR	5.5	5.2	5.5	6.2	6.5	6.4	6.2	6.1	5.8	5.5	5.4	5.4
		MCDBR	5.6	4.9	6.1	8.0	9.2	8.4	8.4	7.3	6.5	5.4	4.8	4.6
		MCWBR	6.4	5.0	4.7	5.5	6.6	6.3	6.4	5.3	5.5	5.3	5.7	5.7
		MCWBR	5.6	4.7	5.1	6.9	8.8	7.6	7.5	6.2	5.6	5.2	5.0	4.7
(35) Mean Daily Temperature Range	5% WB	MCWBR	6.6	5.1	4.5	5.1	7.1	6.2	6.0	5.5	5.5	5.8	5.9	5.7
(40) Clear-Sky Solar Irradiance	taub		0.315	0.338	0.354	0.389	0.376	0.365	0.380	0.385	0.364	0.345	0.319	0.294
	taud		2.206	2.290	2.338	2.291	2.404	2.460	2.437	2.420	2.464	2.404	2.271	2.172
	Ebn at Noon		162	218	251	259	272	277	270	259	247	215	161	132
	Edh at Noon		16	23	29	35	34	32	32	31	26	21	15	12
(44) All-Sky Solar Radiation	RadAvg		108	299	730	1230	1665	1862	1686	1291	837	399	150	75
	RadStd		22	61	75	136	193	167	177	103	84	59	31	10

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)		+0.65	N/A	N/A	N/A	N/A	N/A	-241	-329	+112	N/A

Nomenclature: See separate page